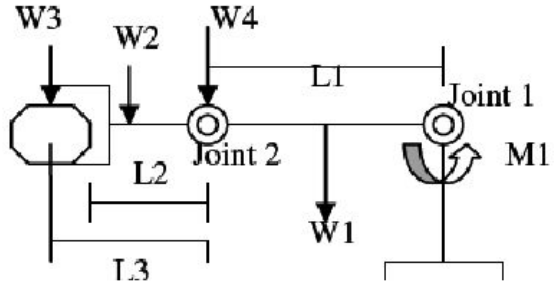


Inverse Dynamics



What's Wrong...



Torque about joint 1:

$$M1 = \frac{L1}{2} * W1 + L1 * W4 + (L1 + \frac{L2}{2}) * W2 + (L1 + L3) * W3$$

Torque about joint 2:

$$M2 = \frac{L2}{2} * W2 + L3 * W3$$

Solid estimate... but does not consider ALL affects...

Rigid-body equations of motion

applied joint torque

$$\tau = M(q)\ddot{q} + C(q, \dot{q})\dot{q} + g(q)$$

joint acceleration

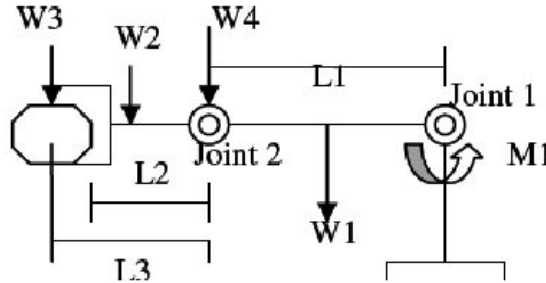
joint velocity

■ aka the **inverse** dynamics

→ maps motion to torque $(q, \dot{q}, \ddot{q}) \rightarrow \tau$

Amazing estimate, parametric. Considers all effects and configurations

I disagree... this should be the worst case scenario!



Torque about joint 1:

$$M1 = \frac{L1}{2} * W1 + L1 * W4 + (L1 + \frac{L2}{2}) * W2 + (L1 + L3) * W3$$

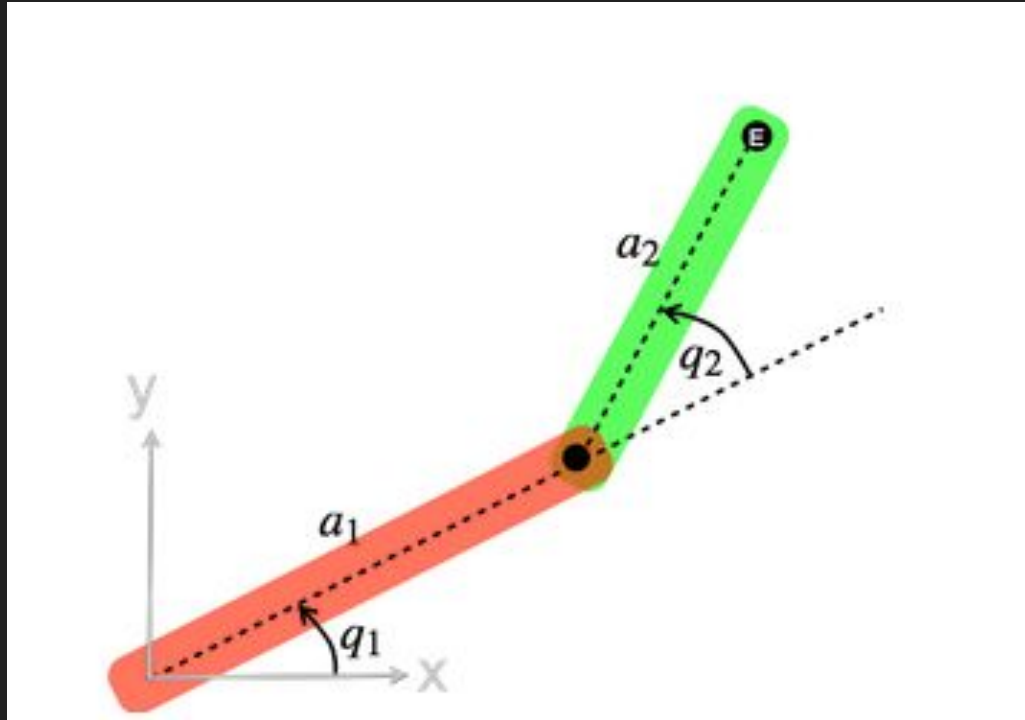
Torque about joint 2:

$$M2 = \frac{L2}{2} * W2 + L3 * W3$$

This is actually fine... if you want to move terribly **SLOW!**

Dynamic effects are not **ALWAYS** negligible, and are especially important for electrical spec'ing.

Generalized Coordinates and Linear Algebra

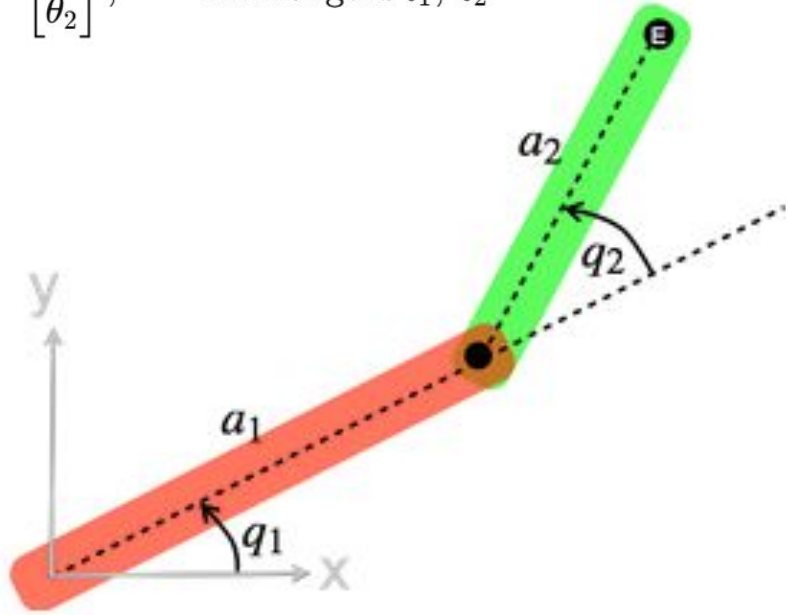


Revolute Revolute Robot Arm

Position and angle of the end effector can be described by two variables...

Generalized Coordinates and Linear Algebra

$$\mathbf{q} = \begin{bmatrix} \theta_1 \\ \theta_2 \end{bmatrix}, \quad \text{link lengths } l_1, l_2$$



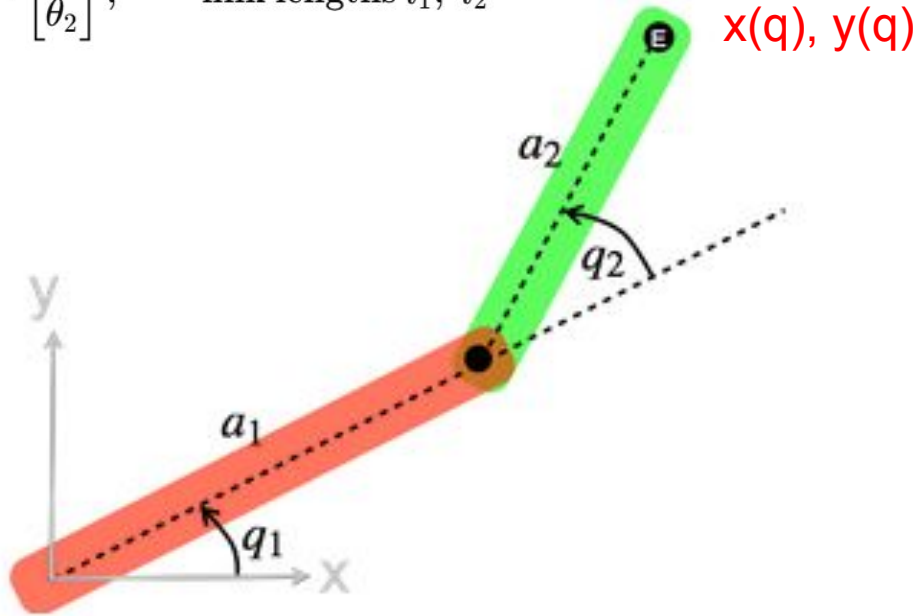
Revolute Revolute Robot Arm

Position and angle of the end effector can be described by two variables...

q_1 and q_2 !

Generalized Coordinates and Linear Algebra

$$\mathbf{q} = \begin{bmatrix} \theta_1 \\ \theta_2 \end{bmatrix}, \quad \text{link lengths } l_1, l_2$$

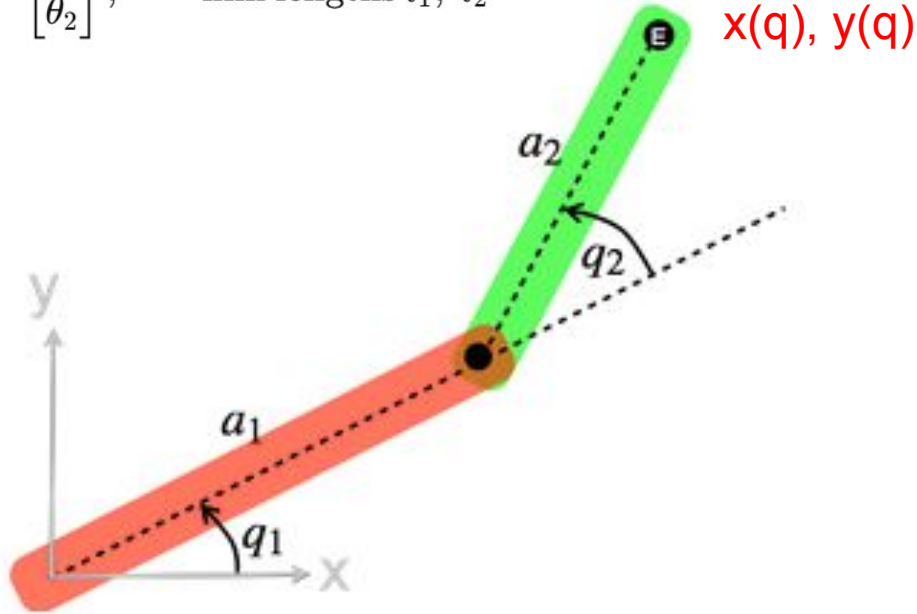


$$\mathbf{p}(\mathbf{q}) = [x(\mathbf{q}), y(\mathbf{q})]^T$$

What is the end effect position X and Y as a function of our now generalized coordinates!

Generalized Coordinates and Linear Algebra

$$\mathbf{q} = \begin{bmatrix} \theta_1 \\ \theta_2 \end{bmatrix}, \quad \text{link lengths } l_1, l_2$$



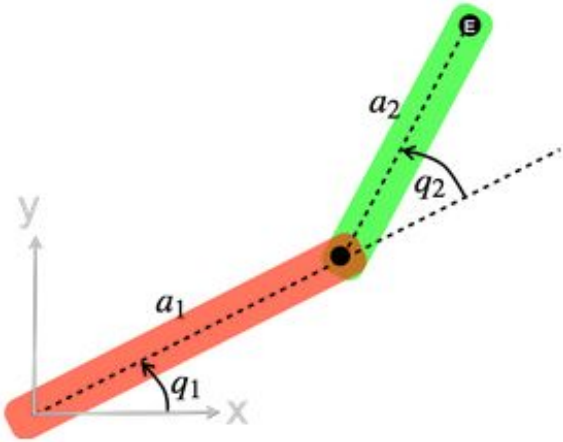
$$\mathbf{p}(\mathbf{q}) = [x(\mathbf{q}), y(\mathbf{q})]^T$$

Quick Aside: if this was 3 DOF planar, we would also include:

$$\mathbf{q} = \begin{bmatrix} \theta_1 \\ \theta_2 \\ \theta_3 \end{bmatrix}$$

Generalized Coordinates and Linear Algebra

$$\mathbf{q} = \begin{bmatrix} \theta_1 \\ \theta_2 \end{bmatrix}, \quad \text{link lengths } l_1, l_2$$



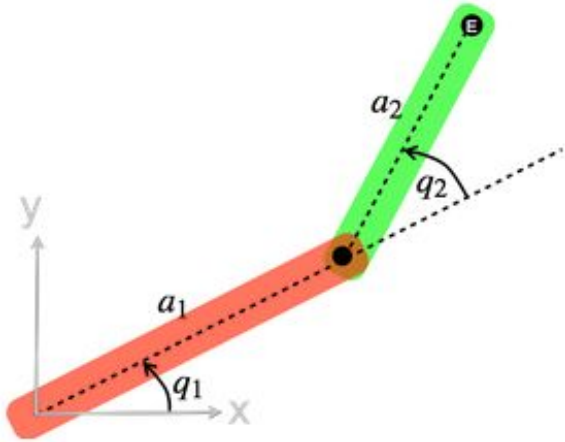
$$x(\theta_1, \theta_2) = l_1 \cos \theta_1 + l_2 \cos(\theta_1 + \theta_2)$$

$$y(\theta_1, \theta_2) = l_1 \sin \theta_1 + l_2 \sin(\theta_1 + \theta_2)$$

$$\mathbf{p}(\mathbf{q}) = \begin{bmatrix} l_1 \cos \theta_1 + l_2 \cos(\theta_1 + \theta_2) \\ l_1 \sin \theta_1 + l_2 \sin(\theta_1 + \theta_2) \end{bmatrix}$$

Generalized Coordinates and Linear Algebra

$$\mathbf{q} = \begin{bmatrix} \theta_1 \\ \theta_2 \end{bmatrix}, \quad \text{link lengths } l_1, l_2$$



$$\dot{\mathbf{p}} = \frac{d}{dt} \mathbf{p}(\mathbf{q})$$

$$\dot{\mathbf{p}} = \frac{\partial \mathbf{p}}{\partial \mathbf{q}} \dot{\mathbf{q}} = \mathbf{J}(\mathbf{q}) \dot{\mathbf{q}}$$

$$\mathbf{J}(\mathbf{q}) = \frac{\partial \mathbf{p}}{\partial \mathbf{q}} = \begin{bmatrix} \frac{\partial x}{\partial \theta_1} & \frac{\partial x}{\partial \theta_2} \\ \frac{\partial y}{\partial \theta_1} & \frac{\partial y}{\partial \theta_2} \end{bmatrix}$$

Generalized Coordinates and Linear Algebra

$$\mathbf{J}(\mathbf{q}) = \frac{\partial \mathbf{p}}{\partial \mathbf{q}} = \begin{bmatrix} \frac{\partial x}{\partial \theta_1} & \frac{\partial x}{\partial \theta_2} \\ \frac{\partial y}{\partial \theta_1} & \frac{\partial y}{\partial \theta_2} \end{bmatrix}$$

**Holy grail of robot
kinematics!**

Generalized Coordinates and Linear Algebra

$$\dot{\mathbf{p}} = \frac{\partial \mathbf{p}}{\partial \mathbf{q}} \dot{\mathbf{q}} = \mathbf{J}(\mathbf{q}) \dot{\mathbf{q}}$$

$$\dot{\mathbf{q}} = \mathbf{J}(\mathbf{q})^{-1} \dot{\mathbf{p}}$$

$$\Delta \mathbf{q} = \mathbf{J}(\mathbf{q})^{-1} \Delta \mathbf{p}$$

NUMERICAL
Inverse
Kinematics

THESE ARE
FOR SMALL
VARIATIONS
OF P

Euler-Lagrange

$$\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}_i} \right) - \frac{\partial L}{\partial q_i} = Q_i$$

$$\text{Lagrangian} = L = \text{KE} - \text{PE} = T - V$$

$$\boldsymbol{\tau} = \begin{bmatrix} \tau_1 \\ \tau_2 \end{bmatrix} = \mathbf{M}(\mathbf{q})\ddot{\mathbf{q}} + \mathbf{C}(\mathbf{q}, \dot{\mathbf{q}})\dot{\mathbf{q}} + \mathbf{g}(\mathbf{q})$$

Inverse Dynamics (NO END EFFECTOR FORCE)

$$\boldsymbol{\tau} = \begin{bmatrix} \tau_1 \\ \tau_2 \end{bmatrix} = \mathbf{M}(\mathbf{q})\ddot{\mathbf{q}} + \mathbf{C}(\mathbf{q}, \dot{\mathbf{q}})\dot{\mathbf{q}} + \mathbf{g}(\mathbf{q})$$

Inverse Dynamics (NO END EFFECTOR FORCE)

$$\boldsymbol{\tau} = \begin{bmatrix} \tau_1 \\ \tau_2 \end{bmatrix} = \mathbf{M}(\mathbf{q})\ddot{\mathbf{q}} + \mathbf{C}(\mathbf{q}, \dot{\mathbf{q}})\dot{\mathbf{q}} + \mathbf{g}(\mathbf{q})$$

Inertia
Matrix

2nd Time
Derivative of
Generalized
Coordinates
Vector

Coriolis /
Centrifugal
Matrix

1st Time
Derivative of
Generalized
Coordinates
Vector

Gravity
Vector

Inverse Dynamics (NO END EFFECTOR FORCE)

$$\boldsymbol{\tau} = \begin{bmatrix} \tau_1 \\ \tau_2 \end{bmatrix} = \mathbf{M}(\mathbf{q})\ddot{\mathbf{q}} + \mathbf{C}(\mathbf{q}, \dot{\mathbf{q}})\dot{\mathbf{q}} + \mathbf{g}(\mathbf{q})$$

Joint
Torques

Inertia
Matrix

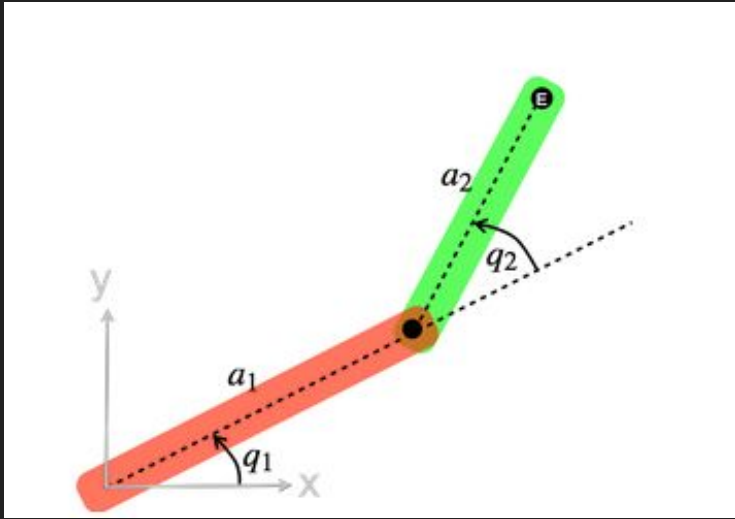
2nd Time
Derivative of
Generalized
Coordinates
Vector

Coriolis /
Centrifugal
Matrix

1st Time
Derivative of
Generalized
Coordinates
Vector

Gravity
Vector

Inverse Dynamics (NO END EFFECTOR FORCE)



Jacobians of COMs

Link 1 COM position

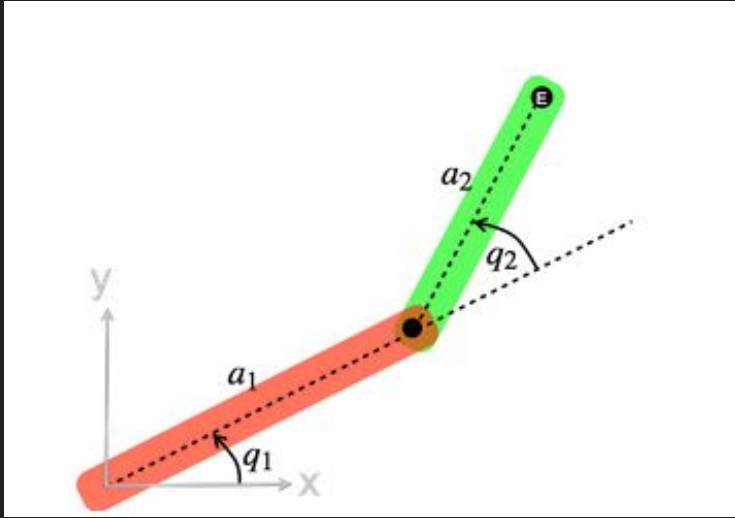
$$p_{c1}(q) = \begin{bmatrix} x_{c1} \\ y_{c1} \end{bmatrix} = \begin{bmatrix} r_1 \cos q_1 \\ r_1 \sin q_1 \end{bmatrix}$$

Link 2 COM position

$$p_{c2}(q) = \begin{bmatrix} x_{c2} \\ y_{c2} \end{bmatrix} = \begin{bmatrix} l_1 \cos q_1 + r_2 \cos(q_1 + q_2) \\ l_1 \sin q_1 + r_2 \sin(q_1 + q_2) \end{bmatrix}$$

$$J_{v,i}(q) = \begin{bmatrix} \frac{\partial x_{ci}}{\partial q_1} & \frac{\partial x_{ci}}{\partial q_2} \\ \frac{\partial y_{ci}}{\partial q_1} & \frac{\partial y_{ci}}{\partial q_2} \end{bmatrix} \cdot$$

Inverse Dynamics (NO END EFFECTOR FORCE)

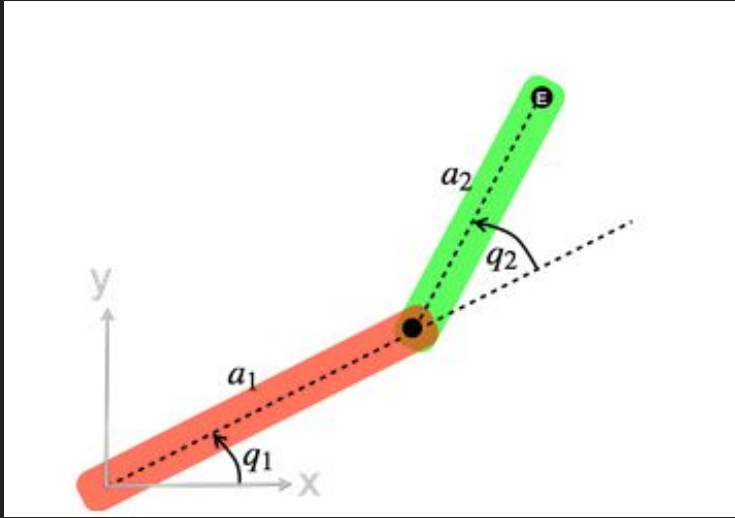


$$J_{v,1} = \begin{bmatrix} -r_1 \sin q_1 & 0 \\ r_1 \cos q_1 & 0 \end{bmatrix}$$

$$\begin{aligned} \frac{\partial x_{c2}}{\partial q_1} &= -l_1 \sin q_1 - r_2 \sin(q_1 + q_2), & \frac{\partial x_{c2}}{\partial q_2} &= -r_2 \sin(q_1 + q_2) \\ \frac{\partial y_{c2}}{\partial q_1} &= l_1 \cos q_1 + r_2 \cos(q_1 + q_2), & \frac{\partial y_{c2}}{\partial q_2} &= r_2 \cos(q_1 + q_2) \end{aligned}$$

$$J_{v,2} = \begin{bmatrix} -l_1 \sin q_1 - r_2 \sin(q_1 + q_2) & -r_2 \sin(q_1 + q_2) \\ l_1 \cos q_1 + r_2 \cos(q_1 + q_2) & r_2 \cos(q_1 + q_2) \end{bmatrix}$$

Inverse Dynamics (NO END EFFECTOR FORCE)



Angular velocity Jacobians $J_{\omega,i}$

Link 1: $\omega_1 = \dot{q}_1$

Link 2: $\omega_2 = \dot{q}_1 + \dot{q}_2$

$$J_{\omega,1} = \begin{bmatrix} 1 & 0 \end{bmatrix}, \quad J_{\omega,2} = \begin{bmatrix} 1 & 1 \end{bmatrix}$$

Inverse Dynamics (NO END EFFECTOR FORCE)

Inertia matrix $M(q)$

$$\dot{p}_{ci} = J_{v,i} \dot{q}$$

$$\omega_i = J_{\omega,i} \dot{q}$$

$$T_i = \frac{1}{2} m_i \dot{p}_{ci}^T \dot{p}_{ci} + \frac{1}{2} I_i \omega_i^2.$$

$$T_i = \frac{1}{2} \dot{q}^T (m_i J_{v,i}^T J_{v,i} + I_i J_{\omega,i}^T J_{\omega,i}) \dot{q}$$

Inverse Dynamics (NO END EFFECTOR FORCE)

Inertia matrix $M(q)$

$$T = \frac{1}{2} \dot{q}^T M(q) \dot{q}, \quad M(q) = \sum_{i=1}^2 (m_i J_{v,i}^T J_{v,i} + I_i J_{\omega,i}^T J_{\omega,i})$$

NICE! ONE TERM COMPLETE!

Inverse Dynamics (NO END EFFECTOR FORCE)

Contribution from link 1

$$J_{v,1}^T J_{v,1} = \begin{bmatrix} r_1^2 (\sin^2 q_1 + \cos^2 q_1) & 0 \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} r_1^2 & 0 \\ 0 & 0 \end{bmatrix}$$

$$m_1 J_{v,1}^T J_{v,1} = \begin{bmatrix} m_1 r_1^2 & 0 \\ 0 & 0 \end{bmatrix}$$

$$J_{\omega,1}^T J_{\omega,1} = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \begin{bmatrix} 1 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} \Rightarrow I_1 J_{\omega,1}^T J_{\omega,1} = \begin{bmatrix} I_1 & 0 \\ 0 & 0 \end{bmatrix}$$

$$M^{(1)} = \begin{bmatrix} I_1 + m_1 r_1^2 & 0 \\ 0 & 0 \end{bmatrix}$$

Inverse Dynamics (NO END EFFECTOR FORCE)

Contribution from link 2

$$J_{v,2}^T J_{v,2} = \begin{bmatrix} l_1^2 + r_2^2 + 2l_1r_2 \cos q_2 & r_2^2 + l_1r_2 \cos q_2 \\ r_2^2 + l_1r_2 \cos q_2 & r_2^2 \end{bmatrix}$$

$$J_{\omega,2}^T J_{\omega,2} = \begin{bmatrix} 1 \\ 1 \end{bmatrix} \begin{bmatrix} 1 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} \Rightarrow I_2 J_{\omega,2}^T J_{\omega,2} = \begin{bmatrix} I_2 & I_2 \\ I_2 & I_2 \end{bmatrix}$$

$$M^{(2)} = \begin{bmatrix} I_2 + m_2(l_1^2 + r_2^2 + 2l_1r_2 \cos q_2) & I_2 + m_2(r_2^2 + l_1r_2 \cos q_2) \\ I_2 + m_2(r_2^2 + l_1r_2 \cos q_2) & I_2 + m_2r_2^2 \end{bmatrix}$$

Inverse Dynamics (NO END EFFECTOR FORCE)

$$M(q) = M^{(1)} + M^{(2)}$$

$$M_{11} = I_1 + I_2 + m_1 r_1^2 + m_2 (l_1^2 + r_2^2 + 2l_1 r_2 \cos q_2)$$

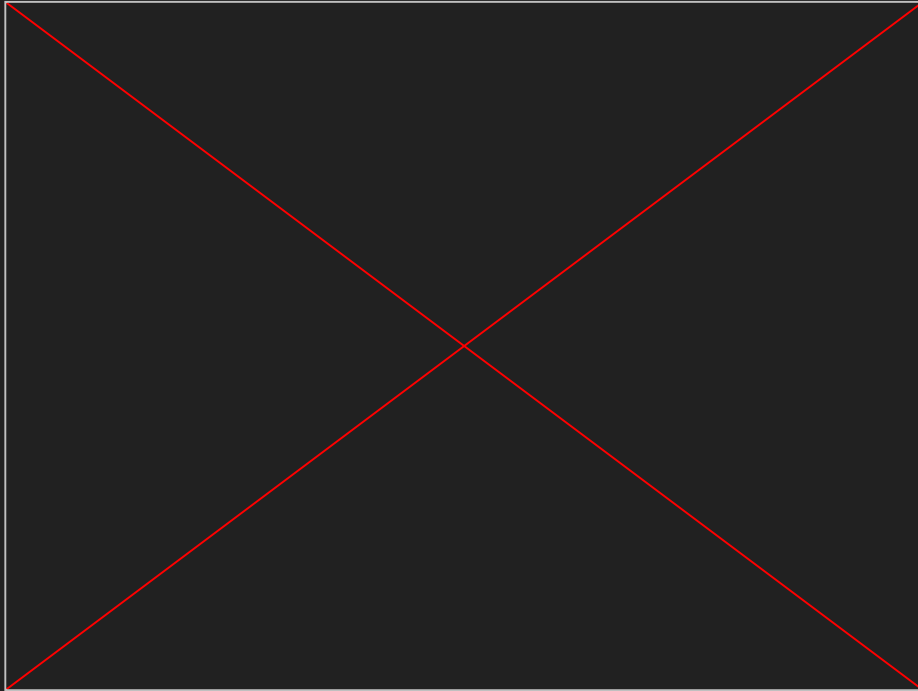
$$M_{12} = I_2 + m_2 (r_2^2 + l_1 r_2 \cos q_2)$$

$$M_{22} = I_2 + m_2 r_2^2$$

NICE! ONE TERM ACTUALLY
COMPLETE!

Inverse Dynamics (NO END EFFECTOR FORCE)

Coriolis/centrifugal matrix



$$C_{ij}(q, \dot{q}) = \sum_{k=1}^2 \Gamma_{ijk}(q) \dot{q}_k$$

Christoffel summations

Inverse Dynamics (NO END EFFECTOR FORCE)

Coriolis/centrifugal matrix

$$\Gamma_{ijk}(q) = \frac{1}{2} \left(\frac{\partial M_{ij}}{\partial q_k} + \frac{\partial M_{ik}}{\partial q_j} - \frac{\partial M_{jk}}{\partial q_i} \right)$$

$$M_{11} = I_1 + I_2 + m_1 r_1^2 + m_2 (l_1^2 + r_2^2 + 2l_1 r_2 \cos q_2)$$

$$M_{12} = I_2 + m_2 (r_2^2 + l_1 r_2 \cos q_2)$$

$$M_{22} = I_2 + m_2 r_2^2$$

Inverse Dynamics (NO END EFFECTOR FORCE)

Coriolis/centrifugal matrix

$$\frac{\partial M_{ij}}{\partial q_1} = 0, \quad \frac{\partial M_{ij}}{\partial q_2} \neq 0 \text{ only for } M_{11}, M_{12}$$

$$\frac{\partial M_{11}}{\partial q_2} = -2m_2 l_1 r_2 \sin q_2$$

$$\frac{\partial M_{12}}{\partial q_2} = -m_2 l_1 r_2 \sin q_2$$

Inverse Dynamics (NO END EFFECTOR FORCE)

Coriolis/centrifugal matrix

$$h(q) = m_2 l_1 r_2 \sin q_2$$

$$\frac{\partial M_{11}}{\partial q_2} = -2h, \quad \frac{\partial M_{12}}{\partial q_2} = -h$$

$$C(q, \dot{q}) = \begin{bmatrix} -h\dot{q}_2 & -h(\dot{q}_1 + \dot{q}_2) \\ h\dot{q}_1 & 0 \end{bmatrix}$$

$$C_{ij}(q, \dot{q}) = \sum_{k=1}^2 \Gamma_{ijk}(q) \dot{q}_k$$

$$\Gamma_{ijk}(q) = \frac{1}{2} \left(\frac{\partial M_{ij}}{\partial q_k} + \frac{\partial M_{ik}}{\partial q_j} - \frac{\partial M_{jk}}{\partial q_i} \right)$$

2nd UNKNOWN TERM!

Inverse Dynamics (NO END EFFECTOR FORCE)

Gravity vector

$$V = m_1 g (r_1 \sin q_1) + m_2 g (l_1 \sin q_1 + r_2 \sin(q_1 + q_2))$$

$$G(q) = \frac{\partial V}{\partial q} = \begin{bmatrix} \frac{\partial V}{\partial q_1} \\ \frac{\partial V}{\partial q_2} \end{bmatrix}$$

$$G_1 = (m_1 r_1 + m_2 l_1) g \cos q_1 + m_2 r_2 g \cos(q_1 + q_2)$$

$$G_2 = m_2 r_2 g \cos(q_1 + q_2)$$

Inverse Dynamics (NO END EFFECTOR FORCE)

$$q = \begin{bmatrix} q_1 \\ q_2 \end{bmatrix}, \quad \dot{q} = \begin{bmatrix} \dot{q}_1 \\ \dot{q}_2 \end{bmatrix}, \quad \ddot{q} = \begin{bmatrix} \ddot{q}_1 \\ \ddot{q}_2 \end{bmatrix}$$

$$\begin{aligned} \tau_1 = & \left[I_1 + I_2 + m_1 r_1^2 + m_2 (l_1^2 + r_2^2 + 2l_1 r_2 \cos q_2) \right] \ddot{q}_1 \\ & + \left[I_2 + m_2 (r_2^2 + l_1 r_2 \cos q_2) \right] \ddot{q}_2 \\ & - (m_2 l_1 r_2 \sin q_2) (2\dot{q}_1 \dot{q}_2 + \dot{q}_2^2) \\ & + (m_1 r_1 + m_2 l_1) g \cos q_1 + m_2 r_2 g \cos(q_1 + q_2) \end{aligned}$$

$$\begin{aligned} \tau_2 = & \left[I_2 + m_2 (r_2^2 + l_1 r_2 \cos q_2) \right] \ddot{q}_1 \\ & + \left[I_2 + m_2 r_2^2 \right] \ddot{q}_2 \\ & + (m_2 l_1 r_2 \sin q_2) \dot{q}_1^2 \\ & + m_2 r_2 g \cos(q_1 + q_2) \end{aligned}$$

Inverse Dynamics... End Effector Force



Inverse Dynamics + End Effector Force



Inverse Dynamics + End Effector Force

$$\mathbf{J}(\mathbf{q}) = \frac{\partial \mathbf{p}}{\partial \mathbf{q}} = \begin{bmatrix} \frac{\partial x}{\partial \theta_1} & \frac{\partial x}{\partial \theta_2} \\ \frac{\partial y}{\partial \theta_1} & \frac{\partial y}{\partial \theta_2} \end{bmatrix}$$

$$\tau_{\text{task}} = J(q)^T F_e$$

or...

$$\tau_{\text{ext}} = J(q)^T F_{\text{ext}}$$

Inverse Dynamics + End Effector Force

$$\tau = M\ddot{q} + b + G + J^T F_{\text{env}}$$

F_{env} = force applied *by the environment on the robot*

$$F_{\text{env}} = \begin{bmatrix} 0 \\ -5g \end{bmatrix}$$

Inverse Dynamics + End Effector Force

$$\tau = M\ddot{q} + b + G - J^T F_e$$

F_e = force applied *by the robot on the environment*

DON'T WORRY. Just do the first one

Inverse Dynamics Conclusion

So.. is that it?

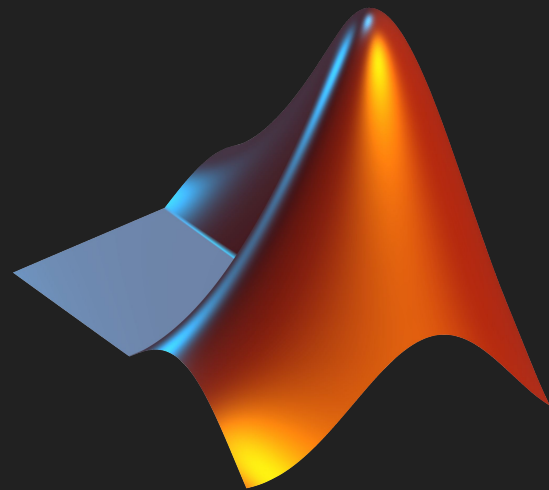
Inverse Dynamics Conclusion

So.. is that it?

$$q = \begin{bmatrix} q_1 \\ q_2 \end{bmatrix}, \quad \dot{q} = \begin{bmatrix} \dot{q}_1 \\ \dot{q}_2 \end{bmatrix}, \quad \ddot{q} = \begin{bmatrix} \ddot{q}_1 \\ \ddot{q}_2 \end{bmatrix}$$

FOR ALL q , $\dot{q}$ AND $\ddot{q}$

So... CONSTRAIN YOURSELF TO
WHERE YOU EXPECT TO
OPERATE



MuJoCo + URDF

